

CS-412 Operating Systems (Minor Test I)

Date: March 04, 2025

Time: 1.15 hr

Marks: 20

Answer all the questions. Each question carries 5 marks.

1. Why is a user mode and kernel mode considered good operating system design? Give an example that illustrate a user process being switched from user mode to kernel mode and then back to user mode.
2. Suppose we are the developers of an operating system that uses a five state process model i.e. new, ready, running, exit and blocked. In our OS, all processes execute in a round robin fashion. Further suppose that we are developing a new operating system for computers that are expected to have numerous I/O bound jobs. To ensure these jobs receive the processor quickly when needed (i.e. when a current I/O burst is finished), we are considering two READY states in our process model: the READY-CPU state (for CPU bound processes) and the READY-IO state (for I/O bound processes). Processes in the READY-IO state will be given the highest priority to access the CPU in the system and will execute in a FCFS fashion. Processes in the READY-CPU state will execute in a round robin fashion. Processes are classified as either CPU-bound or IO-bound when they enter the system.

Discuss the pro/cons of implementing the OS with two ready states as mentioned here.

3. Consider the following five processes to run in a single CPU using preemptive priority scheduling. Show the scheduling order of the processes using Gantt chart. What is the awaiting time for each process and average waiting time?

Process	Arrival Time	Burst Time	Priority
P1	0	3	3
P2	1	4	2
P3	2	6	4
P4	3	4	6
P5	5	2	10

4. What does the term ‘**busy waiting**’ means? Discuss another technique that could be used to avoid busy waiting. Then compare and contrast these two methods.

**School of Computer and Systems Sciences
Jawaharlal Nehru University
New Delhi**

**End Semester Examination
Winter Semester 2025**

CS412--Operating Systems

Time: 3 Hrs. Date: May 22, 2025 Marks: 60

Note: Answer all the questions. Each question carries (7.5 + 7.5 = 15) marks.

1. (a) System calls represent the primary interface between user applications and the kernel. How does the OS ensure that system call execution is atomic? Discuss the performance implications of system call overhead, including context switching, mode switching, and parameter marshalling. Demonstrate with an example.
- (b) How does deadlock avoidance impact system performance? Consider the possible scenario.

scenario.

Given Matrices												
	Allocation Matrix (No of the allocated resources By a process)				Max Matrix Max resources that may be used by a process				Available Matrix Not Allocated Resources			
	A	B	C	D	A	B	C	D	A	B	C	D
P ₀	0	1	1	0	0	2	1	0	1	5	2	0
P ₁	1	2	3	1	1	6	5	2				
P ₂	1	3	6	5	2	3	6	6				
P ₃	0	6	3	2	0	6	5	2				
P ₄	0	0	1	4	0	6	5	6				
Total	2	12	14	12								

Use the safety algorithm to test if the system is in a safe state or not? If the system is in a safe state, can the following requests be granted, why or why not?

- a. P₁ requests (2,1,1,0)
- b. P₁ requests (0,2,1,0)
2. (a) Race conditions are possible in many computer systems. Consider a banking system that maintains an account balance with two functions: **deposit (amount)** and **withdraw (amount)**. These two functions are passed the amount that is to be deposited or withdrawn from the bank account balance. Assume that a husband and wife share a bank account. Concurrently, the husband calls the withdraw () function and the wife

calls deposit (). Describe how a race condition is possible and what might be done to prevent the race condition from occurring. Write the pseudo code to explain the scenario.

(b) How does the principle of least privilege apply to modern operating system security, and what mechanisms do OS implement to enforce it at the process, kernel, and user levels?

3. (a) Describe how demand paging works and how it helps manage memory effectively. Explain the role of page faults in demand paging and how the OS handles them.

(b) Explain the concept of internal and external fragmentation in memory allocation. Discuss how different memory allocation techniques can lead to fragmentation and how to mitigate it.

4. (a) How can different disk scheduling algorithms be chosen to optimize for different I/O patterns? Suppose that a disk drive has 500 cylinders, numbered 0 to 4999. The drive is currently serving a request at cylinder 2150 and the previous request was at cylinder 1805. The queue of pending requests, in FIFO order, is:

2069, 1212, 2296, 2800, 544, 1618, 356, 1523, 4965, 3681

Starting from the current head position, what is the total distance (in cylinders) that the disk arm moves to satisfy all the pending requests for each of the following disk scheduling algorithms:

- SCAN 7402
- SSTF 7398

(b) Explain how file systems implement access control mechanisms, such as access control lists (ACLs) and capabilities. Consider a file system in which a file can be deleted and its disk space reclaimed while links to that file still exist. What problems may occur if a new file is created in the same storage area or with the same absolute path name? How can these problems be avoided?